

Name:

ID #:

Instructor: Maia Martcheva

Directions: You have 2 hours to answer the following questions. You must show all your work as neatly and clearly as possible and indicate the final answer clearly. You may not use a calculator.

Problem	Possible	Points
1	20	
2	20	
3	20	
4	20	
5	20	
Total	100	

(1) (20 points) Show that the fixed point equation $x = f(x)$ with a fixed point α can be solved iteratively, if $f'(\alpha) \neq 1$, by one of the following fixed point iteration formulas:

(a) $x_{n+1} = f(x_n)$

(b) $x_{n+1} = f^{-1}(x_n)$.

Hint: The following formula for the derivative of an inverse is valid

$$[f^{-1}]'(x) = \frac{1}{f'(f^{-1}(x))}$$

(c) $x_{n+1} = (x_n + f(x_n))/2$

(2) (20 points) Let $f(x) \in C[a, b]$. Let $p(x)$ be a polynomial for which

$$\|f' - p\|_{\infty} \leq \epsilon$$

and define

$$q(x) = f(a) + \int_0^x p(t) dt, \quad a \leq x \leq b.$$

Show that $q(x)$ is a polynomial that satisfies

$$\|f - q\|_{\infty} \leq \epsilon(b - a)$$

(3) (20 points) Consider the integral

(1)
$$\int_0^\pi x^2 \cos x \, dx$$

(a) Consider a quadrature rule of the form

$$\int_0^\pi x^\alpha f(x) \, dx \approx Af(0) + B \int_0^\pi f(x) \, dx$$

where $\alpha > -1, \alpha \neq 0$ is a parameter. Determine the constants A, B so that the quadrature formula has degree of exactness one.

(b) Use the formula in part (a) to approximate the integral (1).

- (4) (20 points) For the function $f(x) = \ln(1+x)$ for $x \in [0, 1]$, find the minimax approximation polynomial of degree one. Give the exact value of the minimax error.

(5) (20 points) Assume that you are solving the initial value problem

$$\begin{aligned}y' &= f(t, y) & a \leq t \leq b \\ y(a) &= \alpha\end{aligned}$$

- (a) Derive the formula for the global error of the numerical solutions for the ODE problem above obtained via Euler's method.

Hint: The formula is ($M = \|Y''\|_\infty$):

$$|Y(t_i) - w_i| < \frac{hM}{2L} [e^{L(b-a)} - 1].$$

- (b) Compute the value of $M = \|Y''\|_\infty$ necessary to apply the global error formula above to the specific ODE problem

$$\begin{aligned}y' &= \sin(t + 2y) + e^t & 0 \leq t \leq 1 \\ y(0) &= 0\end{aligned}$$